

A Customized Particle Swarm Method to Solve Highway Alignment Optimization Problem

Yousef Shafahi & Mehdi Bagherian*

Department of Civil Engineering, Sharif University of Technology, Tehran, Iran

Abstract: *Optimizing highway alignment requires a versatile set of cost functions and an efficient search method to achieve the best design. Because of numerous highway design considerations, this issue is classified as a constrained problem. Moreover, because of the infinite number of possible solutions for the problem and the continuous search space, highway alignment optimization is a complex problem.*

In this study, a customized particle swarm optimization algorithm was used to search for a near-optimal highway alignment, which is a compound of several tangents, consisting of circular (for horizontal design) and parabolic (for vertical alignment) curves. The selected highway alignment should meet the constraints of highway design while minimizing total cost as the objective function. The model uses geographical information system (GIS) maps as an efficient and fast way to calculate right-of-way costs, earthwork costs, and any other spatial information and constraints that should be implemented in the design process. The efficiency of the algorithm was verified through a case study using an artificial map as the study region. Finally, we applied the algorithm to a real-world example and the results were compared with the alignment found by traditional methods.

1 INTRODUCTION

The goal of highway alignment optimization is to find the route between two given points that is cheapest to build (Jong and Schonfeld, 2003). To find a convenient approach to this problem, it should be noted that there are an infinite number of possible routes connecting two points and that a complex nonlinear cost function must be identified. Moreover, numerous constraints, such as highway design criteria and impassible regions,

put these issues in the realm of constrained problems. These specific conditions may make it impossible to find the exact solutions that will result in a near-optimal answer using metaheuristic approaches.

Traditional highway design methods have typically been based on the experiences of engineers and the evaluation of a limited number of variants for the selection of the best among them. Because of the large number of possible paths, it is unlikely that such a rudimentary method can lead to a near-optimal solution. In addition, highway design is usually a time-consuming process for the designers. From these shortcomings and the unavailability of precise solutions, metaheuristics has emerged as an appropriate approach to such a complex constrained problem.

Numerous studies in the literature of highway alignment can be found. Most research has addressed the optimization of vertical alignment (Easa, 1988; Goh et al., 1988; Fwa, 1989; Revelle et al., 1996; Jong, 1998; Chan and Sim, 2002; Jong and Schonfeld, 2003), and some other studies have attempted to find the optimum horizontal path (OECD, 1973; Parker, 1977; Wan, 1995; Thomson and Sykes, 1988; Jong, 1998; Jong et al., 2000). However, fewer studies have been directed toward three-dimensional highway alignment optimization. Chew et al. (1989) was the first to propose a three-dimensional model of highway alignment optimization. Using a simple cubic spline polynomial, their model is not a real-world condition because of its simplicity and its violation of highway design criteria. Other models, such as those by De Smith (2006) and Cheng and Lee (2006) have also been proposed for three-dimensional highway alignment optimization.

The most comprehensive studies in the area of highway design automation have been conducted by the University of Maryland's research group. The group's ongoing studies, initiated by Jong (1998), have been advancing over the past decade. Jong and Schonfeld used a metaheuristics optimization algorithm based on a

*To whom correspondence should be addressed. E-mail: info@mehdibagherian.com.

genetic algorithm (GA), a classic search method that depends on the survival of a best answer through successive generations and is utilized to solve a large number of engineering problems (e.g., Adeli and Cheng, 1993; Jiang and Adeli, 2008; Al-Bazi and Dawood, 2010; Marano et al., 2011; Baraldi et al., 2011; Putha et al., 2012). Jong defined genes with real numbers and eight distinct mutations and crossover operators (uniform, straight, nonuniform, whole nonuniform mutations, simple, two-point, arithmetic, and heuristic crossover) that were developed to find new solutions (Jong and Schonfeld, 2003). To give it the ability to solve complex real problems, the model was combined with a geographical information system (GIS) (Jha and Schonfeld, 2004). Jha et al. (2006) defined a set of conditions to be met for an acceptable highway alignment optimization model to be considered efficient.

Over the past years, application of swarm-based algorithms in engineering problems has absorbed researcher's attention. Swarm intelligence is a branch of artificial intelligence, which is based on the study of individual's behavior in various decentralized systems (Teodorovic, 2008). Particle swarm optimization (PSO) (Kennedy and Eberhart, 1995) and ant colony optimization (Dorigo et al., 2006) are of the most known swarm-based algorithms. Among them, application of ant colony optimization is studied in numerous researches (e.g., Putha et al., 2012; Mazloumi et al., 2012). PSO, on the other hand, despite its robust potential upon complex engineering problems (e.g., Plevris and Papadrakakis, 2011), is rarely applied to transportation problems.

In this study, a new search method was developed as an alternative for Jong's proposed GA-based algorithm. The new method attempts to search for an optimal route using a customized PSO algorithm. PSO is a powerful method inspired by the social behavior of bird flocking and fish schooling that was first proposed by Kennedy and Eberhart (1995). In addition, a new approach has been used for exploiting the spatial data from a GIS database. Compared with other studies in the field (Cohn et al., 1992; Kang, 2008), this strategy demonstrates comparable precision but is much faster than recent approaches to using GIS-based data in highway alignment optimization.

The next section of this study describes the costs of the used items and various other items that were considered in the final cost function. In the third section, various aspects of the developed program are described. The search method and the proposed operators are explained in the fourth section. In the fifth section, the model is presented. The numerical examples section outlines the results of the proposed program over an artificial example as well as a real-world problem of high-

way alignment design. In the final section, the results are discussed, and the major findings of this research are summarized.

2 HIGHWAY COSTS

The objective function was composed of three main parts: (1) length-dependent costs, including all cost items that could be derived from the length of an alignment; (2) right-of-way costs, which were obtained by the intersection of the alignment with GIS maps; and (3) earthwork cost. These aspects will be reviewed briefly in this section. Additional information can be found in the research of Jong (1998) and Kang (2008) among other works.

2.1 Length-dependent costs

Length-dependent costs, which are the cost of each unit length of an alignment, can be calculated directly by multiplying the unit cost per length in each possible solution. In this study, the costs of users, including user travel time and vehicle operating costs, are also regarded as length-dependent costs. This calculation will be accurate if we assume a constant design speed, although traffic characteristics are known and fixed.

The unit vehicle-operating cost can be estimated by summing (1) the unit consumption costs, calculated by multiplying the fuel consumption by the fuel price, and (2) maintenance and tire costs. The travel time cost is calculated based on the amount of time spent for traveling and the driver's perceived value of time. The model used for the calculation of user costs is described in detail in the research of Kang (2008).

Construction and maintenance costs are the other integral parts of length-dependent costs. The calculation of each element of this part of the cost function is performed using sufficient details based on unit cost of each item that will be used in a highway construction project. The unit length cost is calculated to facilitate further calculations before starting the search process.

2.2 Right-of-way costs

Right-of-way costs can be calculated by the intersection of the possible solution (a candidate alignment) with the right-of-way from a GIS map. To do so, the candidate paths can be converted to a set of points (known as stations), and the right-of-way cost can be calculated by summing the right-of-way costs for each station. In fact, we assumed the affected area for each single station to be homogenous (i.e., fixed unit right-of-way cost).

By presuming this assumption, the distance between stations should be selected in an efficient way to meet the requirements while avoiding excessive calculations. By assuming a fixed width (D) and intervals between successive stations (S), right-of-way costs of a highway (C_N) were formulated as follows:

$$C_N = DS \sum_{i=1}^{st} C_N(i) \quad (1)$$

where $C_N(i)$ is the unit right-of-way cost value at the co-ordination of station i . In the artificial example, a 10-m interval was regarded as the distance between two successive stations.

2.3 Earthwork costs

The earthwork costs of the project include cutting and filling costs as well as carrying the soils from borrow or to waste. We used the ‘‘Average End Area’’ method (Wright, 1996) to calculate cutting and filling volumes. In this approach, the alignment was primarily divided into a set of sections using cutting planes. The three-dimensional coordination of the alignment at each cutting plane was then calculated using an interpolation process. Second, the elevation of the ground for each station point was exploited using GIS files of the region of study’s topography. The indication of difference between the ground and the station elevations exhibits the type of earthwork in that specific section (cut or fill). Second, the area of cutting or filling in each section is obtained. This can be computed using the exploited elevations, the width of the highway, and the cutting and filling slopes. The earthwork cost for each single section can be calculated using the average area of successive cross-sections multiplied by the length of the section. Thus, the total filling and cutting volumes can be calculated as well. The details and formulation of the method is shown in Jong (1998) and Jha et al. (2006) and is not presented in this article.

Once the earthwork operation volumes were obtained, the difference between volumes (E_N) was calculated using the following equation:

$$E_N = E_C \times s_e - E_F \quad (2)$$

where E_C is cutting volume (m^3); E_F is filling volume (m^3); and S_e is the shrinkage factor of the soil.

The earthwork cost can be calculated using the following equation:

$$C_E = K_C E_C + K_F E_F + K_W \max\{E_N, 0\} - K_B \min\{E_N, 0\} \quad (3)$$

where K_C is unit cut cost ($\text{US}\$/\text{m}^3$), K_F is unit fill cost ($\text{US}\$/\text{m}^3$), K_W is cost of moving earth to waste

($\text{US}\$/\text{m}^3$), and K_B is cost of moving earth from borrow ($\text{US}\$/\text{m}^3$).

3 MODEL STRUCTURE

In this section, we discuss various aspects of the proposed methodology. First, the definition of a three-dimensional alignment and the parameters of the problem are clarified. Second, the proposed coordinate system is described. Third, a concise overview of the implemented constraint handling approaches is provided. Finally, incorporation of GIS maps in the optimization process is outlined.

3.1 Definition of a three-dimensional alignment

The method used in this study is a general form of the model presented in Jong and Schonfeld (2003). In this model, each possible route can be defined by a series of intersection points (PIs) that are connected to the start and end points in each head. In other words, these are the variables of the problem of highway design optimization. PIs include both points of vertical intersections (PVIs) and horizontal intersections (PHIs). Each PI is only a PVI (PHI) when the horizontal (vertical) curvature is zero at that point. A point can be considered to be both horizontal and vertical points of intersection when neither of its curvatures is zero. After the routes are defined, a three-dimensional alignment can be determined by the insertion of a circular curve (for horizontal alignment) and a parabolic curve (for a vertical path) on each intersection. This process was performed using the curve fitting algorithm presented in Jha et al. (2006). The definition of the horizontal and vertical curves would be conducted based on AASHTO (2001) design criteria. The radius of horizontal and length of vertical curves are not the variables of the optimization algorithm. In fact for each set of PIs, they are calculated using the input parameters and geometric design criteria. The minimum radius (which is a function of design speed, maximum superelevation, and side friction of the highway) are used as the horizontal radius. Besides, the length of vertical curves is calculated using the sight distance and algebraic difference of grades in each point of vertical curve. Further details and proposed equations are presented in AASHTO (2001). In this study, spiral curves were not considered along the path because the model and algorithm aims at finding the preliminary design of highways. In addition, because structures are not considered in this study, sight distance at undercrossing criterion for sag vertical curves was not utilized as a design constraint.

In the previous studies the PIs were bounded to a few number of planes, named cutting planes (Jong, 1998; Kang et al., 2009) and the highway alignment optimization problem was solved using a GA-based discrete searching algorithm. The model proposed in this study allows the PIs to locate everywhere in the study area. In other words, the proposed model here tries to find the best alignment in a continuous space without any limitation.

3.2 Coordinate system

To facilitate calculations, we used a local coordinate system in which the origin point $(0, 0, Z_s)$ coincides with the start point and the X -axis is along the line connecting the start and end points. Because the spatial information should be obtained from a map with a general coordinate system, two-dimensional transformation should be applied. Spatial information is necessary for two sets of data: topography information to calculate the earthwork costs and right-of-way information is necessary to calculate the location-dependent costs. In each case, for the required precision, the alignment is converted to a set of three-dimensional station points. Each set is then transformed to a real coordinate system, and the information is derived directly from available maps.

3.3 Constraint handling

When designing a highway alignment, in addition to achieving the minimum cost, it is necessary to meet design criteria and avoid crossing prohibited areas. In this study, according to AASHTO design guidelines (2001), three constraints were considered that must be met for an alignment to be considered acceptable. These constraints are (1) a minimum radius of the horizontal curves, (2) a maximum grade for each section in a profile map, and (3) a minimum length for vertical curves. These requirements were based on the AASHTO design guidelines and were integrated with the model with the aid of a static penalty function. In this method, for each violation, a constant penalty cost and a value proportional to the degree of violation is applied to the overall cost.

The values of the implemented parameters of the penalty cost's functions should be selected such that they neither cause abrupt convergence to a local condition nor are so low that they might violate the optimization process. These values should be calibrated to gain the best efficiency of the algorithm.

Crossing infeasible areas can be avoided by using a large unit right-of-way cost for the area (Jha et al., 2006). When an alignment crosses the area, it incurs a large cost and decreases its chances to be considered

the best and most economical alignment design. In addition, initial solutions are selected inside the study area, although it is possible for some answers to lie outside of the study area. This quality can be accomplished by the inertia term of the proposed PSO algorithm, which is discussed in Section 4.

Finally, the crossing solution modifier (which will be described in the next section) can be regarded as a method to omit a number of impossible solutions and replace them with better ones.

3.4 Dealing with infeasible areas

One of the most common reasons for trapping in local optimum may emerge from the right-of-way cost component. This phenomenon is especially noticeable when the infeasible region is large relative to the entire search space. Using GA as the searching algorithm, mutation operators may prevent trapping. However, because the mutation operator is not implemented in the PSO algorithm, an efficient strategy to escape from local optimum seems to be more important for this swarm-based search method. This approach is more critical when velocity clamping is applied in PSO and large displacements of particles are limited to maximum velocity (V_{max}) values and may decrease the searching ability to escape from local optima. Note that the candidate solutions would be evaluated based on total cost and that the model is blind to perceiving each component cost separately.

Right-of-way costs are among the major components of a highway project. There may be numerous areas in each region where passing through is not allowed. Assigning a high right-of-way unit cost is a common strategy to avoid passing through such infeasible areas. Passing through areas with high right-of-way costs would increase the total highway cost dramatically; thus, violating such an area will not be selected as the best alignment.

The inability to improve fitness while passing through a region with uniform right-of-way costs is a major reason for the premature convergence of the searching process. Assume an infeasible region (e.g., a lake) in the search study (Figure 1a). The region can be regarded as having a rather high unit cost to prevent the passing alignment from being considered the best one. Although it may help to compare the alignment with paths that do not pass through the lake, it is unable to distinguish alignments that fall within the lake. Regarding length-dependent cost as another component of the total cost, the proposed alignment may even incline the searching process to opt for the worst of the possible routes, thus distracting the algorithm from exploring further options and causing a premature convergence.

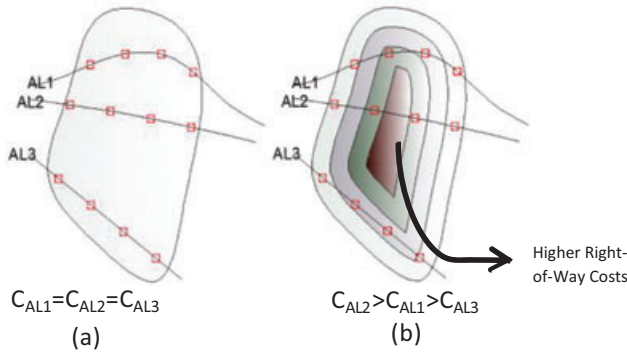


Fig. 1. Schematics of the common methodology of addressing infeasible areas (a) and a heuristic method of considering infeasible areas in a highway alignment optimization model (b).

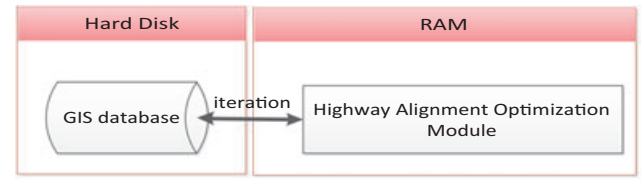
In this study, a heuristic method has been used to improve the efficiency of the algorithm. In this method, the paths are directed to escape from entrapment in local optimum created by infeasible regions. It will try to steer the swarm to the margins of an infeasible region. The proposed method does not regard a uniform value for the whole region, and right-of-way cost has been defined based on distance from the boundaries of the infeasible region. This process is accomplished by buffering the region's boundary toward its center and assumes different unit costs for each ring (Figure 1b). In this case, the algorithm would try to direct particles (candidate solutions) outside of the infeasible region.

3.5 Incorporating a GIS

A GIS is a common method to store spatial data and attributes such as land boundaries and topographic maps. Providing a direct link from GIS maps to an optimization algorithm may be mandatory when solving a real-world highway alignment optimization problem (Jha and Schoenfeld, 2004). Regardless, few studies have incorporated GIS and an optimization model simultaneously.

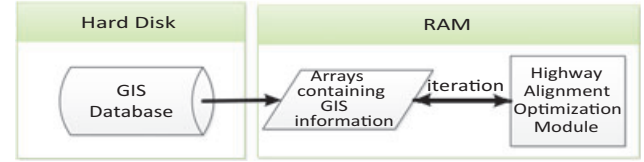
Jha and Schoenfeld (2004) were the first to use a GIS database combined with a GA-based search method. In their study, a programming interface and a GIS environment were linked together using an ArcView Dynamic Link Library called Avhelp.dll. Using this architecture, the computational time for each iteration was more than 90 seconds (Jha et al., 2006). Regarding the iterative nature of a metaheuristic optimization approach, a highway alignment optimization process would take a rather long time using this approach. Kang et al. (2009) states that most of the model's evaluation time (99.98%) is devoted to the right-of-way costs and estimation of the environmental impacts, which are computed by the cus-

Common method of using GIS data in HAO



(a)

Modified method to increase the exploitation speed



(b)

Fig. 2. Common (a) and proposed (b) methods for linking GIS databases to a highway alignment optimization (HAO) module.

tomized GIS module. He proposed a “prescreening and repairing” method to modify the infeasible solutions before sending them to the GIS evaluation module.

In this study, a new method is proposed that exploits GIS databases in a fast and precise way. In this method, using the ArcEngine Library in the .Net environment, each GIS database (that is stored in raster format) would transfer to an array that can store GIS information in the memory of the computer. The process would accomplish this step before starting the search process. The main point is that the transferring process would be conducted only once for each layer using this strategy and, depending on available memory, computation time would be reduced drastically. Each evaluation of a possible alignment would take only a fraction of a second. Figure 2 shows a graphical scheme of the method for exploiting GIS databases. It is worthwhile to note that with this method, the precision of spatial information is dependent on the size of the raster cells (each cell has a single value as the unit right-of-way cost or elevation), and the utilized method does not intervene in the map's precision. In other words, in the proposed approach, the model utilized whatever was stored in a GIS database. Nevertheless, there may be several limitations in the size and precision of the data, which is imported to computer's memory. Assimilation of the current computers with large capacity of available memory is not a major concern. For instance, for the artificial example, 206 megabytes of the memory was reserved to store a $6,600 \times 4,500$ m region with the cell size of 1×1 m where the data were reserved with double precision.

4 SEARCHING METHODOLOGY

In this section, a short overview of the basic model of PSO will be presented. Next, the proposed method of

initializing the model will be discussed. Finally, two proposed operators that were considered in this program are introduced.

4.1 PSO method

Inspired by the social behavior of bird flocking and fish schooling, PSO is an evolutionary computation model that has its roots in artificial life. First proposed by Kennedy and Eberhart (1995), PSO performs a population-based search using particles to represent potential solutions within the search space. Each particle is characterized by its position, velocity, and a record of its past performance.

In the basic PSO algorithm, using the following equations, the position (x) and velocity (v) of each particle in the swarm can be updated after each iteration.

$$x(t+1) = x(t) + v(t+1) \quad (4)$$

$$v(t+1) = v(t) + C_1 r_1(t) [g(t) - x(t)] + C_2 r_2(t) [p(t) - x(t)] \quad (5)$$

where C_1 and C_2 are the accelerator constants, r_1 and r_2 are randomly generated numbers that distribute uniformly in the $[0,1]$ interval, g is the best answer found by the population to that point, and p is the best answer found by each particle. Figure 3 shows the flowchart of the PSO algorithm. The operators added to this method are discussed in Section 4.3.

There are many extensions of basic PSO to improve its convergence behavior. Shi and Eberhart (1998) introduced the “inertia weight model,” in which the inertia of the particle (the $v(t)$ term) would be multiplied by a w parameter, known as inertia weight. Consequently, Equation (5) can be rewritten as

$$v(t+1) = wv(t) + C_1 r_1(t) [g(t) - x(t)] + C_2 r_2(t) [p(t) - x(t)] \quad (6)$$

In this equation, the inertia weight can either stay constant or decrease during the execution of the algorithm.

Another modification is velocity clamping (Eberhart et al., 1996). In each iteration, if the calculated displacement of a particle exceeds the specified maximum velocity, it is set to the maximum value. Let $V_{\max,j}$ denote the maximum velocity allowed in dimension j . The particle velocity is then adjusted before the position update using

$$v_{ij}(t+1) = \begin{cases} v'_{ij}(t+1) & \text{if } v'_{ij}(t+1) \leq V_{\max,j} \\ V_{\max,j} & \text{if } v'_{ij}(t+1) \geq V_{\max,j} \end{cases} \quad (7)$$

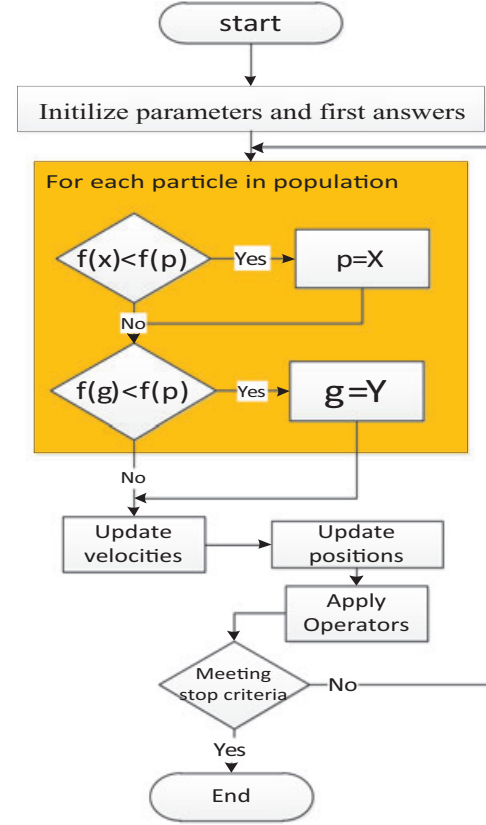


Fig. 3. Schematic of the PSO searching procedure for highway alignment optimization.

where $v'_{ij}(t+1)$ is the speed of the particle, obtained from Equation (6), and $v_{ij}(t+1)$ equals the modified value which is considered $c \ t + 1$ to be the particle's speed at step $t + 1$.

In this study, these equations, in combination with some operators (which will be described in the next section), have been used for a swarm-based search to find the best alignment. For each point of intersection, the sum of three variables is the dimension of the problem. Implementation of the PSO algorithm to solve highway alignment optimization problem is appropriate for two reasons. First, PSO is inherently designed to solve continuous problems and its efficiency is reported in the previous studies. Second, on the contrary of the other proposed algorithms for highway alignment optimization (e.g., GA), the PSO implicitly considers the spatial relations during the searching process. Consequently, this algorithm can be considered as a promising method for alignment optimization problems. These reasons motivated us to develop a PSO-based searching methodology and evaluate its performance.

4.2 Initial solutions

The assignment of a swarm's initial values is the first step in a PSO-based optimization model. This step is important because the efficiency of the PSO can be affected by the initial diversity of the swarm. The positions of the particles are usually initialized using a uniform distribution of particles to uniformly cover the search space. It is critical to cover all of the possible regions of the search space; otherwise, the PSO may have difficulty finding the global best answer because the momentum of PSO is the only way to move particles to uncovered areas (Engelbrecht, 2007). In other words, assuming a specific dimension of a PI (say its elevation), if the initial answers lie between a and b , the momentum of the particle is the only term which lets the algorithm search the values outside the $[a, b]$ interval.

In this study, several methods were used to derive initial answers. Depending on the study area's characteristics and complexity, an appropriate method (or even a mixed set of methods) may be chosen.

To initialize the particle positions, the set of variables was divided into horizontal and vertical parameters. The latter set is assigned using the following relationship:

$$Z(i) = \frac{L(i)}{L_{SE}} \times (Z_E - Z_S) + r(i) \quad (8)$$

where $Z(i)$ = elevation of PI(i); $L(i)$ is the Euclidean distance of point PI(i) from the start point; L_{SE} is the Euclidean distance of the start and end points; and Z_S and Z_E are the elevation of the start and end points, respectively. Finally, $r(i)$ is a random value, uniformly distributed between a and b (i.e., $r(i) \sim U(a, b)$). Actually, using Equation (8), a straight line with some noise would be assigned as the initial elevation of each PI.

The initialization of the horizontal coordinates of the initial alignment has been conducted using various methods. In all of the used procedures, a feasible domain of selection is defined through cutting lines and the study area's boundary. The limitation of the study area to certain cutting lines (and cutting planes for three-dimensional alignments) was primarily proposed by Jong (1998). In this study, the proposed method was used to derive initial answers. By knowing the distances between the cutting lines, the $X(i)$ value for each point of intersection can be easily obtained. Hence, the position of each PI on the corresponding cutting line (i.e., $Y(i)$) is the only remaining variable that should be initialized. Using an arbitrary coordination system (which was described in the third section), the feasible range for each point of intersection is the sum of the left and right distances for each cutting line. It is common to inscribe a search space with a rectangle and assign a high value

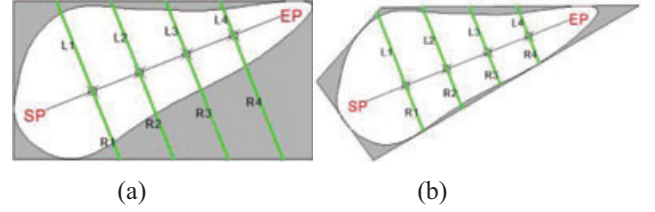


Fig. 4. Inscription of the study area in previous studies (a) and the proposed method using a quadrilateral (b) to decrease the infeasible area.

for the infeasible regions that are within the bands (Jha et al., 2006). However, in this study, a quadrangle (which is the generic form of a rectangle) was used to reduce the infeasible regions and hence decrease the number of initial solutions that may lie outside of the search domain. The hatched region in Figure 4 shows that the proposed method may efficiently narrow the search space. It is noteworthy that we aim at narrowing the cutting lines, regardless of the changes in the area of the study region. To find an appropriate quadrilateral for the problem, a rectangle where its sides were tangent to the region of study was primarily generated in CAD. The vertexes of this polygon were then moved to reach as less infeasible area as possible.

The method used to define the boundary is based entirely on geometric relations. As a result, the upper and lower values for each cutting line can be calculated, and the initial values can be selected within each band. Thus, the following four methods were used to assign the third variable of each point of intersection, $Y(i)$:

1. Initial answers meet the horizontal design criteria (i.e., a minimum radius of the curve between each tangent).
2. $Y(i)$ is assigned using a uniform distribution (i.e., $Y(i) \sim U(L(i), U(i))$, where $L(i)$ and $U(i)$ are lower and upper bound for cutting line i , respectively).
3. For the deviation from the center, in which a triangular distribution is used, initial solutions would be inclined toward the outer parts of study area.
4. Manual inputs let users insert their own initial solutions based on their assumptions and judgment. This function has been provided using a graphical interface that is linked to the highway alignment optimization module.

4.3 Operators

Because of the numerous variables in the highway alignment optimization, it may not be efficient to use the general form of an optimization algorithm. Hence, the search algorithm should be customized for this specific

problem to achieve better solutions with less iteration. In this study, two operators were applied to improve the results of the optimization process: straight mutation and twist modifying.

4.3.1 Straight mutation. The straight mutation operator was applied to guide the searching process to find alignments with shorter length. In this process, the coordination of a selected PI is changed to the average value of its back and forth PI. The use of this operator is motivated by the reliance of a large portion of the objective function value on length-dependent costs, and it is logical to expect a straight section to have a lower cost than a curve. It should be noted that this operator would apply to a small amount of the population and is similar to a mutation in a GA. Moreover, in each application, the operator would modify only one dimension of each PI to avoid a conflict between the vertical and horizontal positions of the candidate point.

Removing a PI is accomplished by the following equation:

$$\begin{cases} x_{i,j}(t) = \frac{x_{i,j-1}(t) + x_{i,j+1}(t)}{2} & \text{if } r < p \\ x_{i,j}(t) = x_{i,j}(t) & \text{if } r \geq p \end{cases} \quad (9)$$

where $x_{i,j}(t)$ is the i th dimension of the j th particle in the t th step, $r \sim u(0, 1)$ is a random value with uniform distribution, and $p \sim [0, 1]$ is the proportion of applying this operator to the entire population. In other words, the probability of applying this operator for each dimension is 100% in each iteration.

4.3.2 Twist modifying. Using a three-dimensional search space (i.e., without considering perpendicular cutting planes) will allow each point of intersection to be placed in each location of the search space. However, this freedom should be carefully controlled to avoid alignment with the intersection itself, named as “twist.” This problem is particularly common during the first iterations of optimization, in which large displacement is possible. It is worth mentioning that in the real-world highway design projects there are a number of situations in which it is not possible to avoid self-intersection of the highway. Inner loop connections are the quintessence of this case. In this study, we have not considered interchanges mainly because of their structural issues and intricate considerations.

To address the problem of highway alignment optimization, each point of intersection is controlled in each step of the algorithm to ensure no intersection with the possible solution. For each PI, the impossible area is calculated first, and if the PI location falls within this area, it is exchanged with the next PI. Figure 5 shows an example in which the impossible area for PI3 (where PI3

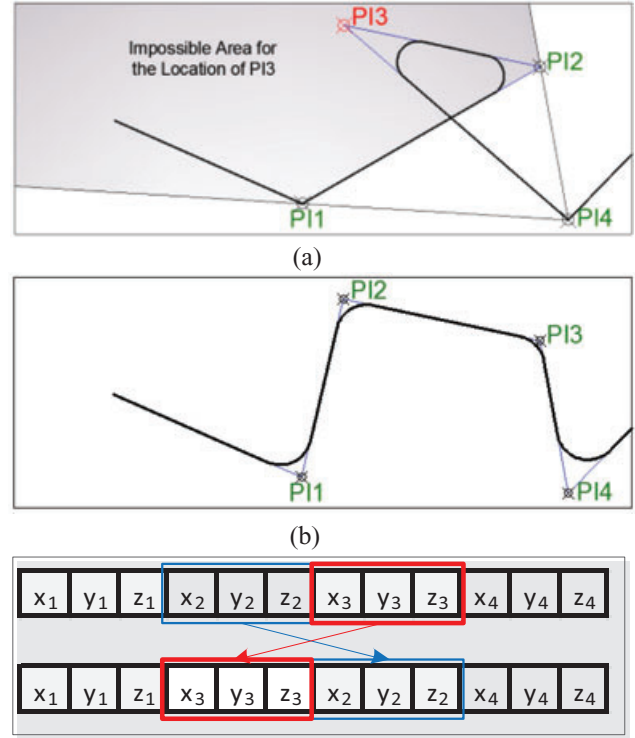


Fig. 5. Proposed twisting modification to replace poor answers (a) with acceptable ones (b).

cannot be located) is calculated, and because this point is placed in that area, its coordinate is exchanged by PI2. For each PI, the boundaries of the impossible area can be obtained by calculation of the three lines in which each pair of points would be collinear.

The applied algorithm for this operator uses the following two basic conditions for each point of intersection to determine whether this point lies within its impossible area. If this condition meets, the successive PIs would be exchanged to solve the twist.

1. The angular condition should be considered, which means that the bearing angle in the PI4–PI3 direction should lie between PI4–PI2 and PI4–PI1.
2. The line connecting PI3 and PI4 must intersect the PI1–PI2 line, which can be verified by checking whether the Euclidean distance between PI3 and PI4 is larger than the distance from PI4 to the intersection point of the lines PI1–PI2 and PI3–PI4.

Using this strategy, the program avoids self-intersecting. It is worthwhile to note that the self-intersection problem was observed for only two successive points. In other words, the clamping of the displacement of particles in each iteration (using maximum velocity) avoids large displacements, even in the initial steps of the optimization process, and such an

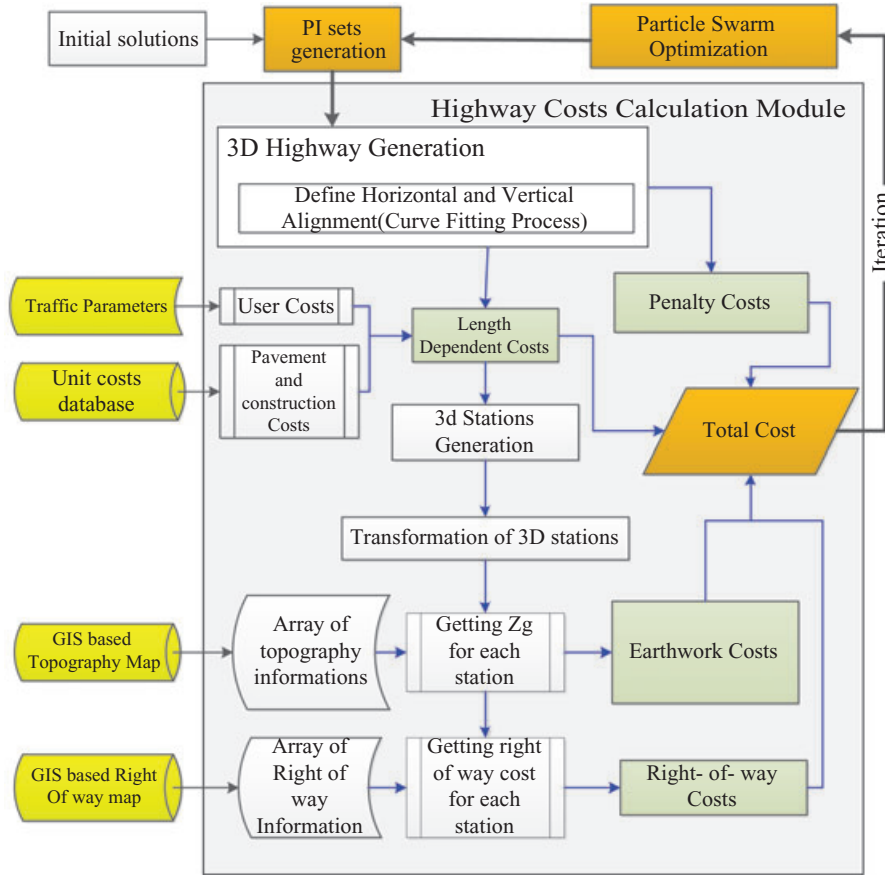


Fig. 6. Structure of the developed program for highway alignment optimization using customized PSO.

occurrence would stop the intricate twisting of possible solutions. To deal with the twist of the PIs, which are not adjacent, a more intricate approach is necessary. Because in the example studies such an issue was not observed, we did not extend the algorithm for the general form. Besides, such alignments are probably inferior because of their excessive length; thus they would be simply rejected over the successive iterations.

5 SUMMARIZED MODEL

Regarding the variables and cost function of a possible answer, it is possible to assess the fitness of each proposed alignment. Using customized PSO method, the best alignment (an alignment with the lowest cost that design criteria have been regarded in) may be obtained. The following is the final model that has been considered in this study.

$$\text{Minimize}_{X_{PI1}, Y_{PI1}, Z_{PI1}, \dots, X_{PIIn}, Y_{PIIn}, Z_{PIIn}} C_T = C_L + C_N + C_E + C_R + C_G + C_M$$

where C_T is the total cost, C_L is the length-dependent cost (including construction and user length dependent costs), C_N is the location-dependent (right-of-way) cost, C_E is the earthwork costs, and C_R , C_G , and C_M are the penalty costs of violating the minimum radius, maximum allowable grade, and minimum allowable vertical length in each PI, respectively. There is no limitation for the coordinates of PIs, and the definition of the initial solution within the possible area and the high cost of the right-of-way for the area outside the study region can ensure that an alignment is found that is not outside the area. Figure 6 provides an overview of the different elements of cost function and their relationship to the search algorithm.

6 NUMERICAL EXAMPLES

In this section, we applied the developed algorithm to two different examples; one in an artificial map and the other a real-world problem. The efficiency of the algorithm for different parameters as well as the efficacy of the operators are evaluated through the example study.

Table 1

Input variables and parameters of the example study

<i>Length-dependent costs</i>	
Length-dependent construction costs	156
Travel time unit costs (US\$/unit length)	73
User costs	32
Total length dependent costs (US\$/m)	261
<i>Earthwork costs</i>	
Cut cost (US\$/m ³)	2.2
Fill cost (US\$/m ³)	3.6
Cut-fill angles (degree)	45
Width (m)	12
<i>Design criteria</i>	
V_{Design} (km/h)	75
Maximum superelevation rate (percent)	8
Coefficient of side friction	0.16
Distance between stations (m)	10
Maximum allowable grade (%)	8

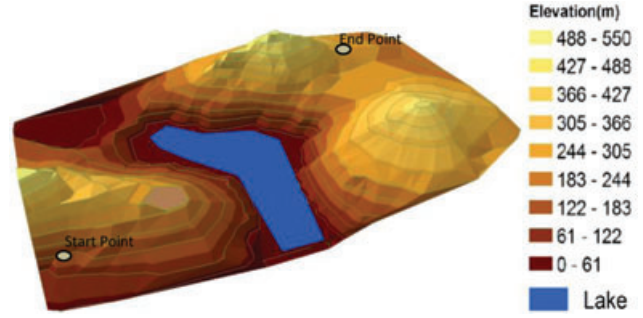
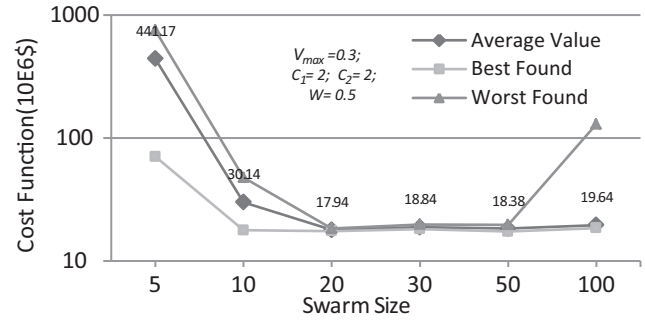
In the second example, a real-world rural highway alignment design problem is solved and the results are compared with what is proposed by the consulting firm using traditional methods.

6.1 An artificial example

To examine the efficiency of the proposed algorithm, an artificial example was employed. The basic design of this example is similar to the example of Jong and Schonfeld (2003). The study region is a $6,600 \times 4,500$ m rectangle and contains three mountains and a lake (which is impossible to cross). In this area, it is impossible to consider a direct path between the start point (300, 1500) and the end point (5450, 4850), and hence the optimum route may be a backtracking one. It is assumed that the elevations of the start and end points are equal to the ground elevation at each point. Moreover, in addition to the lake, three regions with high right-of-way costs were considered as impossible areas. The unit costs and design criteria considered for this example are summarized in Table 1.

The information about the topography and right-of-way costs was provided in a GIS format (i.e., IMG files). The structure of the model is such that it can be used to directly exploit the required information at each stage. Figure 7 shows a perspective view of the study region.

6.1.1 Optimum coefficients of the search method. As mentioned, the optimal value for the inertia weight is problem dependent. To ensure the best handling of the search method and obtain the best possible solution, we should try to identify the best parameters. In this sec-

**Fig. 7.** A perspective view of the study area.**Fig. 8.** Sensitivity analysis on the swarm's population size.

tion, we attempt to find the best values for inertia weight (w), maximum velocity (V_{max}), and the accelerator coefficients (C_1 , C_2).

Swarm population: In this study, a global best (gbest) PSO model was developed for a highway alignment optimization problem, and each particle had a direct link to others through which it communicated (i.e., its own evaluation of the current position) with all of the particles of the swarm.

The size of the swarm depends on the complexity of the search space, and it is recommended that a size in the range of 10–30 be used in common problems (Van den Bergh and Engelbrecht, 2001). To confirm that recommendation, a simple analogy was conducted using six different values. For each size of the population, five independent executions were run. Figure 8 shows the results of these runs. Based on the results, the swarm sizes of 20, 30, and 50 provided more reliable answers. Among them, a swarm size of 20 resulted in the best average of objective function. Consequently, the swarm size of 20 was proposed as an acceptable value for this specific problem.

Maximum velocity: Velocity clamping limits the displacement of each PI in a highway alignment optimization problem. Small possible displacement causes slow convergence of the algorithm and may avoid escaping

from local optima. However, very large displacements may decrease the efficiency of the algorithm.

It is critical to use an appropriate value for the maximum velocity because it can control the behavior of the algorithm. Although large values of $V_{\max}$ facilitate global exploration and allow the algorithm to search for feasible regions all over the study area, smaller values encourage local exploitation and improve the algorithm's local search ability (Engelbrecht, 2007). Nevertheless, there are some threats for the search process in each case. On the one hand, large values of possible displacement may increase the probability of the searching algorithm missing good regions and cause abrupt displacement and twisting of the solutions. On the other hand, small values may increase the number of time steps required to reach an optimum. Furthermore, the swarm may become trapped in a local optimum when its displacements are limited to inappropriate values (Engelbrecht, 2007). If $V_{\max}$ is too small, the swarm may not explore sufficiently beyond local good regions (Eberhart et al., 1996).

In this study, observations show that using a maximum velocity greater than 0.3 of the range of each dimension leads to a divergence from the algorithm. It should be noted that this value is different for each dimension and that each value uses its own range.

Inertia weight: It is important to assign an appropriate value for inertia weight (w) because it can assure that the algorithm is convergent. Moreover, this parameter plays an important role in trading off the exploration and exploitation of the algorithm. Similar to velocity clamping values, large amounts of inertia weight will improve the exploration, but the diversity may also increase. In contrast, small values of w increase the possibility of searching in a specific area to obtain better solutions but may also increase the probability of trapping in local optimums (Parsopoulos and Vrahatis, 2010).

The value of the inertia weight depends on the nature of the problem. Van den Bergh and Engelbrecht (2006) showed that the following equation will derive a convergent algorithm.

$$w > \frac{1}{2}(C_1 + C_2) - 1 \quad (10)$$

In addition, some studies have suggested a range of 0.4–0.9 as a recommended domain for the inertia weight (Engelbrecht, 2007).

To obtain better coefficients, five different values were tested to find the best coefficient as an optimum value. For each value, 10 executions (with different seeds for random variables of the program) were conducted. Moreover, maximum velocity, accelerator coefficients and swarm size were assumed to be constant for

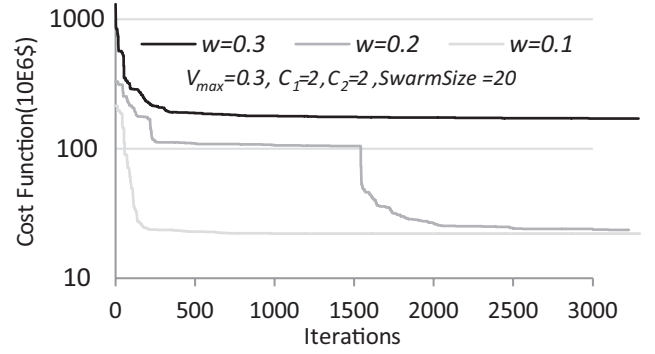


Fig. 9. Changes in total cost in successive iterations for different inertia weights.

each run ($V_{\max} = 0.3$, $C_1 = C_2 = 2$ and swarm size of 20). It was observed that using an inertia weight that is larger than 0.3 cannot yield logical answers, and the PSO used would thus be somehow inefficient. However, despite the relatively large deviation of the results in different runs (likely attributable to trapping in local answers), using $w = 0.1$ results in the best answers compared with other values. Figure 9 compares the best execution for each value of w in 3,000 iterations.

Eberhart and Shi (1998) proposed a linearly decreasing inertia weight as a strategy to use both exploration and exploitation and thus improve the efficiency of the algorithm. In this approach, in each step (t), the inertia weight (w_t) can be computed using the following equation:

$$w_t = w_i - \frac{(w_f - w_i) \times t}{T} \quad (11)$$

where w_i and w_f are the initial and final weights, respectively, and T is total number of iterations.

This approach is used in the example study of a highway alignment optimization program to observe whether it leads to better answers (i.e., alignments with fewer costs). To do so, using $w_i = 0.6$ and $w_f = 0.1$, w_t was decreased using Equation 11. Figure 10 shows that using a dynamic inertia weight may include better solutions in this problem. It can be also observed that the best solution was obtained using the static inertia method, whereas the initial solutions were relatively few when a dynamic weight was started with an inappropriate initial solution. The range of inertia weight chosen was based on trial and error (for the upper bound) and previous results of static weight (for the lower bound). Numerous methods have been proposed for changing inertia weight values, which may be proven in future studies.

Accelerator coefficient: Four different sets of accelerator coefficient values were observed to determine

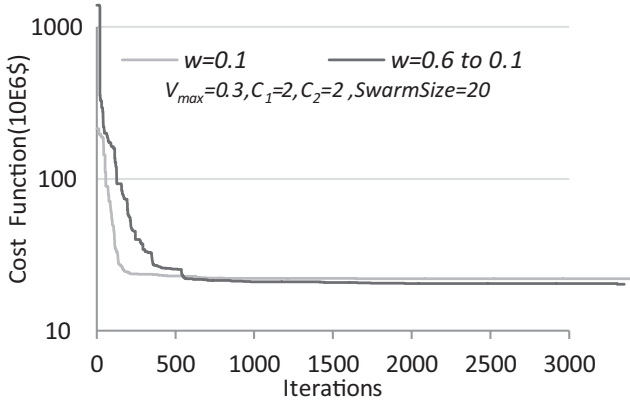


Fig. 10. Total cost changes in successive iterations using constant and dynamic inertia weight.

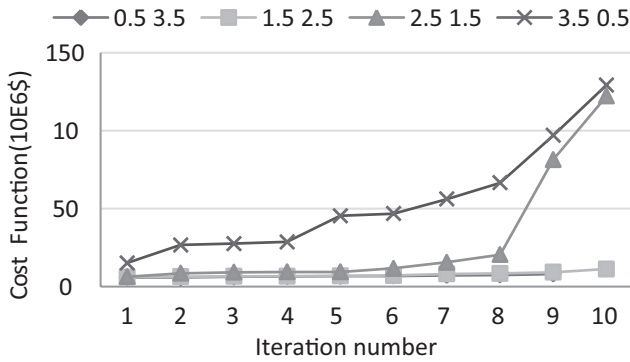


Fig. 11. Sensitivity analysis on different values of accelerator coefficients (C_1 , C_2).

which would lead to a better solution. The acceleration coefficients, C_1 and C_2 , together with the random vectors, r_1 and r_2 , control the stochastic influence of the cognitive and social components on the overall velocity of a particle. Although C_1 controls the effect of the best solution found by each particle, the C_2 value is the weight of the best solution found by a swarm so far.

Figure 11 shows the results of running the model with different values of C_1 and C_2 in 10 independent runs. Results show that using a C_2 value greater than C_1 may result in better solutions for this example. Note that these values cannot be regarded as optimal for each example, and each example may have its own optimum coefficients.

6.1.2 Efficiency of the heuristic operators. To address the performance of the heuristic operators (i.e., straight mutation and twist modification), we made an analogy between the simple PSO and the customized algorithm. To accomplish this task, the problem was solved in four scenarios to evaluate the full algorithm and each single operator separately. As can be seen in Figure 12

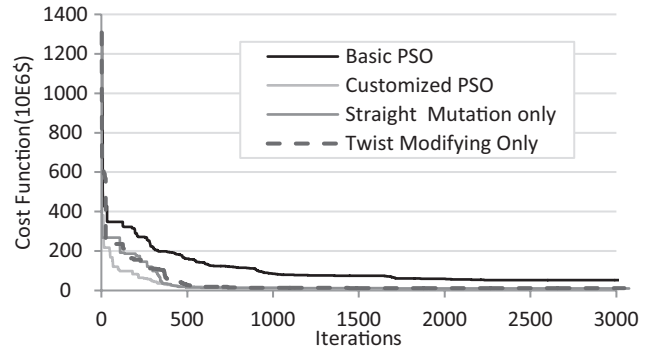


Fig. 12. The efficiency of the heuristic operators on the behavior of the basic PSO.

Table 2

Results of successive executions of a searching method for the case study

Number of run	Objective value $\times 10^6$ (US\$)	Objective value/mean	Computation time (seconds)
1	6.344	0.961	257
2	7.197	1.090	334
3	5.891	0.893	180
4	6.0757	0.921	260
5	6.614	1.002	415
6	6.370	0.965	172
7	6.410	0.971	213
8	6.612	1.002	248
9	7.274	1.102	559
10	7.208	1.092	346
Mean	6.599	1	298
Std	0.484	0.07	—

applying the operators to a certain extent has changed the trend of the algorithm to find the optimum answer. In fact, they not only have improved the trend of the algorithm (especially upon the first iterations), an improvement in finding the best answer was achieved as well.

6.1.3 Computational results. Table 2 shows the results of running the model with the best parameters found in 10 successive and independent runs. Each single run is performed using different seed numbers for random number generations, which are utilized to make initial solutions and stochastic PSO parameters. These runs were performed on a laptop computer with an Intel® Core™ 2 Dual 2.80-GHz processor and 4 GB of installed memory (RAM). Comparing the normal standard deviations of the results with those of Jong's study shows that the proposed algorithm may find more reliable solutions than the developed algorithm, which is based on a GA. However, because of the different unit

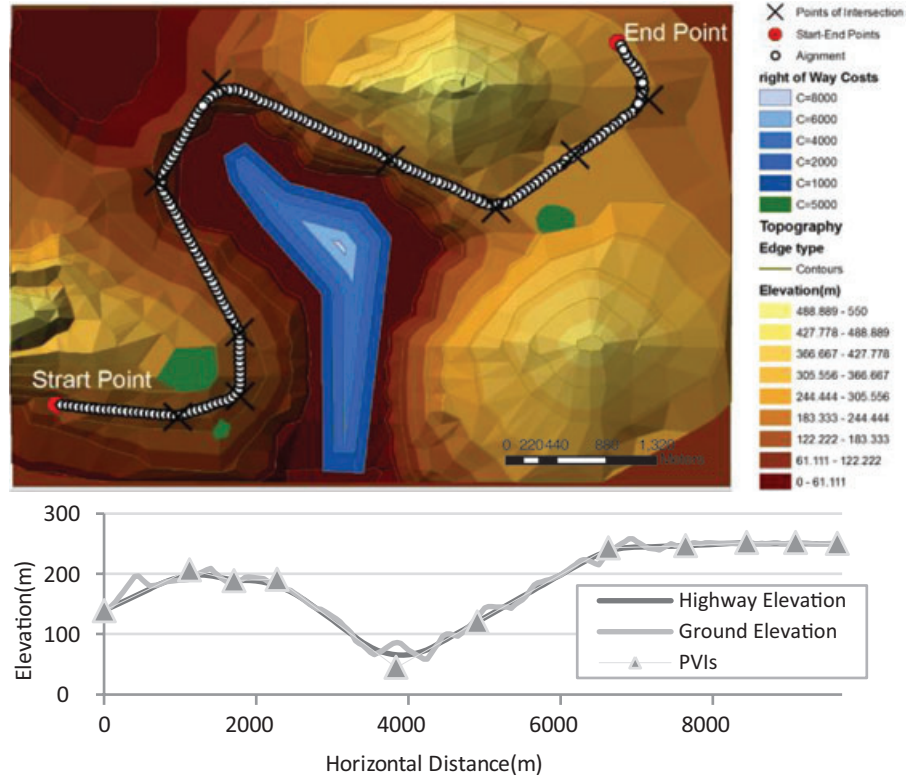


Fig. 13. Plan and profile view of the best solution.

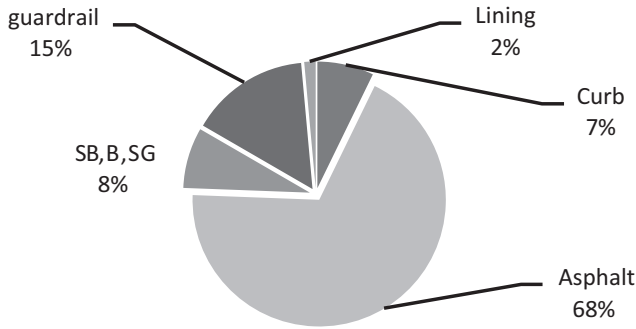


Fig. 14. Percentage of different elements of length-dependent construction costs for the case study example.

costs and different maps used, the quality of best alignment found did not compare with Jong's results. It is noteworthy that we do not know what the exact optimal solution is and the best-found answer by the customized PSO algorithm is assumed to be the best answer. The plan of the best alignment found in the study area and its profile view are shown in Figure 13.

The results further reveal that the efficiency of the developed PSO is dependent on the parameters of the algorithm. In fact, the selection of appropriate parameters

Table 3
The parameters used for unit user costs and travel time calculations

Analysis period (years)	30
Annual average daily traffic (AADT)	100
Interest rate (%)	3
Traffic composition factor	0.05
Fuel cost (US\$/L)	0.4
Auto travel time unit cost (US\$/h)	9
Truck travel time unit cost (US\$/h)	16
Design speed (km/h)	75

is a crucial part of utilizing PSO in highway alignment optimization. The example study indicated that the assigned higher values for the social parameter (C_1) rather than the cognitive one (C_2) resulted in the best efficiency. The results further suggest that a swarm size of 20–50 can lead to more reliable results in this problem. Further tests and applying the model to examples with different characteristics will confirm these suggestions.

Although the variances in the different runs are improved, high variances are still observed in the solution. One strategy to tackle the issue is to repeat the runs to reach the superior answer. Following this

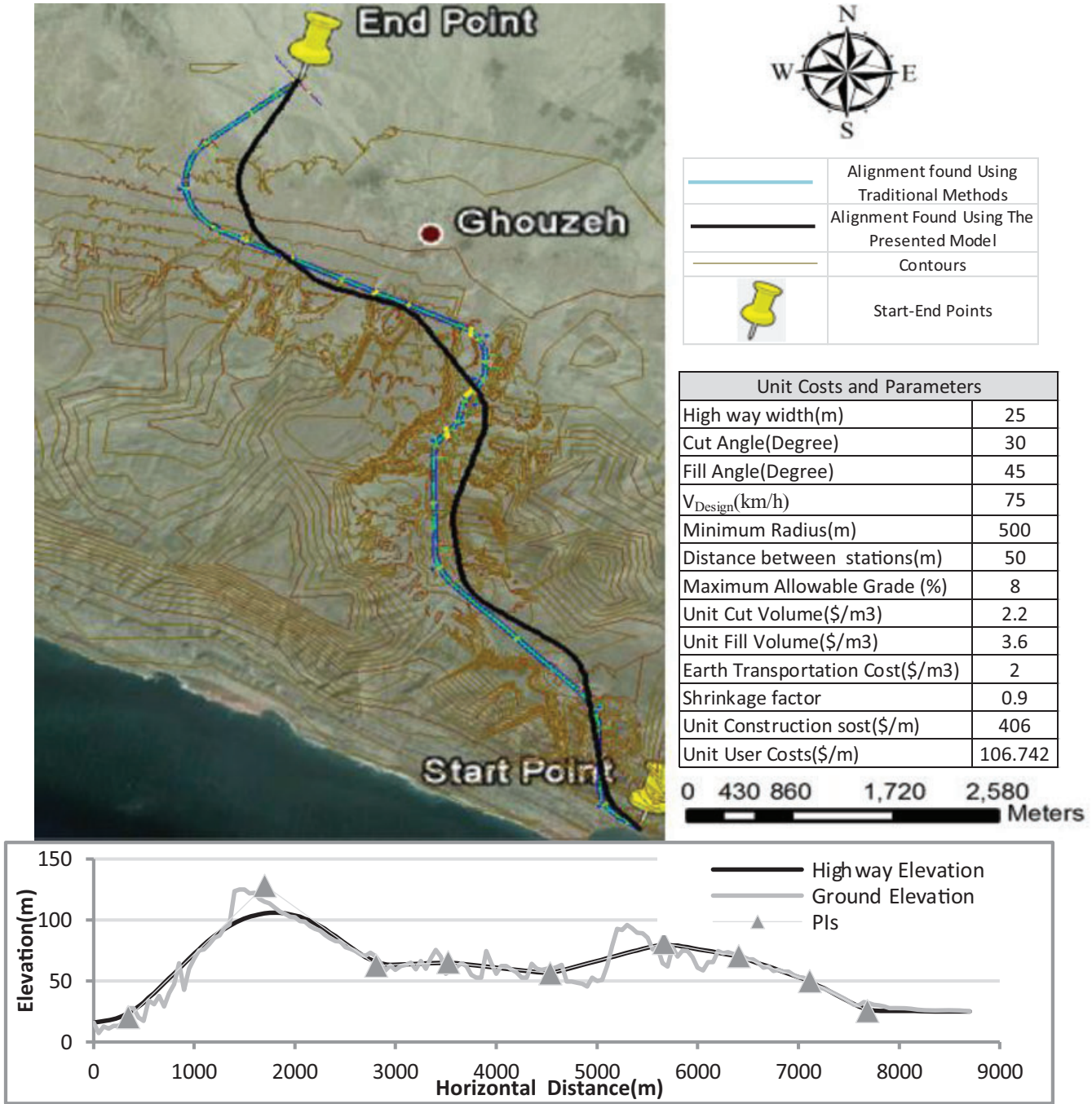


Fig. 15. Plan and profile view of best solution for the case study.

approach, fast computations may be of great importance. Applying the method presented in Section 3.5 is what we implemented to decrease the CPU time. The execution time of the searching process is in the range of 172–559 seconds with an average of 298 seconds for 10 successive runs started with different initial solutions. This is much faster than the results reported in the previous studies, which were trying to have a

link to the GIS database. For instance, the method presented by Kang et al. (2009) named “prescreening and repairing” solved a similar problem (with up to seven numbers of PIs) in 4,077 seconds. This remarkable time saving value shows the robust potential of the method implemented in this study. It is noteworthy that in addition to changing the data exploitation method, we have further made simplifications for right-of-way costs

calculation in which the unit cost value for each single station can be used to calculate that section. In fact, the method presented by (Jong et al., 2000) which is implemented in their other studies performs additional calculations (e.g., spatial analysis and calculation of affected parcels), which needs much computation time compared to what is performed in this study. However, we believe that the formulations described in Section 2.2 are sufficient for the preliminary design of highways, especially for rural regions.

6.2 A real-world case study

We applied the developed model to solve a real-world example and compared the results with what was obtained using traditional methods. The project was defined as designing of a highway, starting from Parsian Port (689280, 3007014) to Ghevarzeh (686515, 3014009). The width of the highway was supposed to be 25 m and the city named “Ghouzeh” was the region the highway could not pass through (i.e., infeasible area). The elevation of the region varied from 0 to 210 m. The topography data were available in contours with 10-m intervals. To exploit them, the file was converted to a Raster file (IMG format) to be readable by the program. During the optimization process, nearly 800 MB of the memory were devoted to storing the topography and right-of-way maps. Finally, the distance between each of the stations was assumed to be 50 m because there were no limitations on right-of-way cost and this value was assumed to be sufficient for volumetric calculations.

The unit construction cost was obtained from Iran’s Highway and Airport unit costs. Figure 14 depicts the calculated portion of each element for the suggested pavement design in Iran, in which the thickness of the hot mix asphalt (HMA), the base layer, and sub base are assumed to be 0.15 m.

Furthermore, a preliminary calculation was performed to calculate the unit user and travel costs, which

are the other length-dependent cost items. As aforementioned, these calculations were performed using the comprehensive formulation presented in Kang (2008). The assumed traffic data are summarized in Table 3.

The optimization process was then performed for this problem. Figure 15 shows the obtained horizontal and vertical alignment which was found by the algorithm. In this figure, the alignment found using traditional methods is shown as well. It can be seen that the alignment pattern is similar to what is designed by traditional methods. Because a significant portion of the cost function was devoted to the length dependent costs, the optimal answer obtained from the model is shorter compared to the alignment found by traditional methods.

Table 4 summarizes the result obtained for each single alignment. As can be seen, a significant improvement in the total cost is obtained using the developed model. However, this model is a preliminary method and can be used as a tool for practitioners and there are many more justifications and points, which should be considered over the design process.

7 CONCLUSION

In this study, a new method was proposed for a highway alignment design problem to minimize costs, including construction costs, user costs, and right-of-way costs. Construction costs were calculated based on real costs of a road construction project and the real values. The optimization algorithm used relies on a customized PSO method that was integrated in a GIS. A new approach to link GIS to highway alignment optimization module was proposed. This allowed the exploitation of information from GIS maps within a fraction of a second and greatly improved the overall execution time. A flexible approach was used to assign initial solutions that allowed the user to choose the most appropriate initializing options from those provided or to manually select the options. The efficiency of the algorithm was verified with a case study using an artificial map. Finally, the model was applied to a real-world problem and the results were compared to the alignment found by traditional methods. The results show that the designated algorithm is an efficient method to automate the first phase of the design process for optimizing highway alignment.

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Table 4

Results obtained by traditional and developed methods for the case study

<i>Cost item</i>	<i>Developed model</i>	<i>Traditional method</i>
Length of alignment (m)	8,698	9,675
Cut volume (m ³)	1,014,002	1,683,305
Fill volume (m ³)	912,602	1,021,823
Earthwork costs (US\$)	5.516E + 06	6.502E + 06
Construction costs (US\$)	3.532E + 06	3.928E + 06
User and travel time costs (US\$)	9.285E + 05	1.033E + 06
Penalty costs (US\$)	0	0
Total cost (US\$)	9.976E + 06	1.217E + 07

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